Chapter 4. Entropy rates of a stochastic process

<u>Recall</u> AEP

X_i : ~~iid~~ with probability dist. $p(x)$.
stationary ergodic process.

$-\frac{1}{n}\log p(X_1, X_2, \cdots, X_n) \longrightarrow \cancel{H(X)}$ in probability.
$H(\mathcal{X})$ "entropy rate"

$\to$ $p(x_1 \cdot x_2, \cdots, x_n) \approx 2^{-nH(X)}$

$\to$ Typical set $A_\varepsilon^{(n)} := \{(x_1, x_2, \cdots, x_n) \in \mathcal{X}^n \mid 2^{-n(H+\varepsilon)} \le p(x_1, x_2, \cdots, x_n) \le 2^{-n(H-\varepsilon)}\}$

$\to$ Data compression.

Q) How many bits are needed to describe X_i : dependent?

Sec 1638

1. entropy rate, Markov chain.
2. entropy rate of stationary Markov chain
3. " functions of Markov chain
4. 2nd law of thermodynamics

§. Markov chain, entropy rate.

- joint p.d.f $p(x_1, x_2, \cdots, x_n) = Pr(X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n)$ $\forall (x_1, x_2, \cdots, x_n) \in \mathcal{X}^n$, $\forall n$.

Markov chain — discrete stochastic proc. $\{X_n \mid n \in \mathbb{N}\}$ s.t.
discrete

$Pr(X_{n+1} = j \mid X_n = i, X_{n-1}, \cdots, X_1) = Pr(X_{n+1} = j \mid X_n = i)$.

$\to$ $p(x_1, \cdots, x_n) = p(x_1)\, p(x_2|x_1)\, p(x_3|x_2) \cdots p(x_n|x_{n-1})$

stationary process. $Pr(X_1 = x_1, \cdots, X_n = x_n) = Pr(X_{1+l} = x_1, X_{2+l} = x_2, \cdots, X_{n+l} = x_n)$
$\forall l, \forall n$.

time-invariant Markov chain s.t.

$Pr(X_{n+1} = j \mid X_n = i) = Pr(X_2 = j \mid X_1 = i) \overset{\text{def}}{=} P_{ij}$

→ For states $1, 2, \cdots, m$,

Initial dist. & transition matrix determines such process.

$$P = \begin{array}{c|cccccc} & 1 & 2 & \cdots & j & \cdots & m \\ \hline 1 & & & & & & \\ 2 & & & & & & \\ \vdots & & & & & & \\ i & & & & p_{ij} & & \\ \vdots & & & & & & \\ m & & & & & & \end{array}$$

$$\Pr(X_{n+1} = j) = \sum_{k=1}^{m} \Pr(X_n = k)\, P_{kj}$$

irreducible — Markov chain s.t.

$\forall i, j \in \{1, 2, \cdots, m\}$, $\exists n \in \mathbb{N}$ s.t.

$\Pr(X_n = j \mid X_1 = i) > 0$

aperiodic — irreducible Markov chain

let d: largest integer s.t.

$\Pr(X_{n+1} = i \mid X_1 = i) > 0 \Rightarrow n \mid d$ for all i

Then, $d = 1$.

Stationary distribution. — distribution $\mu = (\mu_1, \mu_2 \cdots \mu_m)$ s.t.

$\Pr(X_n = i) = \mu_i$, $\forall i = 1, 2, \cdots, m$

$\Rightarrow \Pr(X_{n+1} = i) = \mu_i$, $\forall i$, $\forall n$

* μ is stationary distribution $\Leftrightarrow$. $\mu = \mu P$

* finite-state Markov chain is irreducible & aperiodic $\Rightarrow$

stationary dist. is unique, $\mu_i = \lim_{n \to \infty} P(X_n = i)$ for any initial dist.

entropy rate. 1. Given $\{X_i\}$: stochastic process,

$$H(\mathcal{X}) := \lim_{n \to \infty} \frac{1}{n} H(X_1, X_2, \cdots, X_n).$$

2. Given $\{X_i\}$: stochastic process,

$$H'(\mathcal{X}) := \lim_{n\to\infty} H(X_n|X_{n-1},\cdots,X_1).$$

ex). typewriter. m symbols, n letters

$$\Rightarrow \quad H(X_1,X_2,\cdots,X_n) = -\sum_{x_1,x_2,\cdots,x_n} p(x_1,\cdots,x_n)\log p(x_1\cdots x_n)$$

$$= -\sum \frac{1}{m^n}\log\frac{1}{m^n}$$

$$= \log m^n$$

$$H(\mathcal{X}) = \lim_{n\to\infty}\frac{\log m^n}{n} = \log m.$$

$$\Rightarrow \quad H(X_n|X_{n-1},\cdots,X_1) = -\sum_{x_1\cdots x_n} p(x_1,\cdots,x_n)\log p(x_n|x_{n-1}\cdots x_1)$$

$$= -\sum\frac{1}{m^n}\log\frac{1}{m}$$

$$= \log m$$

$$H'(\mathcal{X}) = \log m.$$

thm 4.2.1 $\{X_i\}$: stationary stochastic process $\Rightarrow$ $H(\mathcal{X}) = H'(\mathcal{X})$.

pf).
$$H(X_{n+1}|X_n,X_{n-1},\cdots,X_1) \le H(X_{n+1}|X_n\cdots X_2)$$
$$= H(X_n|X_{n-1}\cdots X_1).$$

$H(X_n|X_{n-1},\cdots,X_1)$ is decreasing, nonnegative, so has a limit $H'(\mathcal{X})$.

By chain rule, note that.

$$b_n \Rightarrow \frac{1}{n}H(X_1,X_2,\cdots,X_n) = \frac{1}{n}\sum_{i=1}^{n} H(X_i|X_{i-1},\cdots,X_1) \leftarrow a_i$$

Claim). $a_n \to a$, $b_n = \frac{1}{n}\sum_{i=1}^n a_i$, then $b_n \to a$.

Pf of Claim). $\forall \varepsilon>0$, $\exists N$ s.t. $n\ge N$ implies. $|a_n - a| \le \varepsilon$.

$$|b_n - a| = \left|\frac{1}{n}\sum_{i=1}^n (a_i - a)\right|$$
$$\le \frac{1}{n}\sum_{i=1}^n |a_i - a|$$

$$\leq \frac{1}{n}\sum_{i=1}^{N}|a_i - a| + \frac{1}{n}(n-N)\varepsilon$$

$< 2\varepsilon$ for sufficiently large n.

$\lim_{n\to\infty} \frac{1}{n} H(X_1, X_2 \cdots X_n) = H(\mathcal{X})$ exists and $H(\mathcal{X}) = H'(\mathcal{X})$.

P.

$. Entropy rate of stationary Markov chain.

Suppose $\{X_i\}$: stationary Markov chain.

$$\Rightarrow \quad H(\mathcal{X}) = H'(\mathcal{X}).$$

$$= \lim_{n\to\infty} H(X_n | X_{n-1}, \cdots, X_1).$$

$$= \lim_{n\to\infty} H(X_n | X_{n-1})$$

$$= \lim_{n\to\infty} H(X_2 | X_1)$$

$$= H(X_2 | X_1)$$

let μ: stationary dist. of $\{X_i\}$.

Assume $X_1 \sim \mu$

$$\Rightarrow \quad \underline{H(\mathcal{X})}. = H(X_2|X_1)$$

$$= -\sum_{x_1, x_2} p(x_1, x_2) \log p(x_2|x_1).$$

$$= -\sum_{x_1, x_2} \mu_{x_1} P_{x_1 x_2} \log P_{x_1 x_2}.$$

$$= \underline{-\sum_{i,j} \mu_i P_{ij} \log P_{ij}} \qquad (*)$$

* Same applies to <u>irred & aperiodic Markov chain.</u>

$\Downarrow$

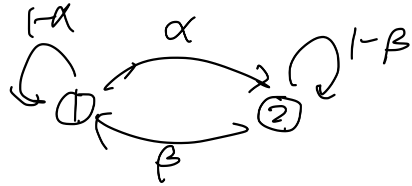
$\exists!$ stationary dist.

$\Downarrow$

Even if $X_1 \not\sim \mu$, $(*)$' holds ex). Consider a Markov chain st

State = $\{1, 2\}$.

$$P = \begin{array}{c c} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \end{array}$$



$\Rightarrow$ let μ : stationary dist.

$$\mu = \mu P$$

$$(\mu_1 \;\; \mu_2) = (\mu_1 \;\; \mu_2) \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}.$$

$$\mu_1 = \mu_1 (1-\alpha) + \mu_2 \beta.$$

$$\begin{cases} \mu_1 \alpha = \mu_2 \beta \\ \mu_1 + \mu_2 = 1 \end{cases}$$

$$\mu_1 = \frac{\beta}{\alpha+\beta}, \quad \mu_2 = \frac{\alpha}{\alpha+\beta}.$$

Suppose $X_1 \sim \mu$. so that $\{X_i\}$ is stationary.

$$\Rightarrow H(\mathcal{X}) = H(X_2 | X_1)$$

$$= -\sum_{i,j} \mu_i P_{ij} \log P_{ij}$$

$$= -\mu_1 \sum_j P_{1j} \log P_{1j} - \mu_2 \sum_j P_{2j} \log P_{2j}$$

$$= \frac{\beta}{\alpha+\beta} H(\alpha) + \frac{\alpha}{\alpha+\beta} \cdot H(\beta).$$

$$\left(H(p) := -p \log p - (1-p) \log (1-p) \right).$$

§ Entropy rate of functions of Markov chains.

Suppose $\{X_i\}$: stationary Markov chain

$Y_i = \phi(X_i)$

$\Rightarrow$ $\{Y_i\}$: not necessarily Markov chain,

but is stationary proc, so $H(\mathcal{Y})$ is well-defined.

Q) What is $H(\mathcal{Y}) = \lim H(Y_n | Y_{n-1}, \cdots, Y_1)$?

Goal : we present two bounds of $H(Y_n | Y_{n-1}, \cdots, Y_1)$.

* $H(Y_n | Y_{n-1} \cdots Y_1)$; decreasing, nonnegative, so it converges.

lem 4.5.1.

$H(Y_n | Y_{n-1}, \cdots, Y_2, X_1) \le H(\mathcal{Y})$.

pf).
$$\begin{aligned} H(Y_n | Y_{n-1}, \cdots, Y_2, X_1) &= H(Y_n | Y_{n-1}, \cdots Y_2, Y_1, X_1) \\ &= H(Y_n | Y_{n-1}, \cdots, Y_2, Y_1, X_1, X_0, X_{-1}, \cdots, X_{-k}) \\ &= H(Y_n | Y_{n-1}, \cdots, Y_2, Y_1, Y_0, \cdots, Y_{-k}, X_1, X_0, \cdots, X_{-k}) \\ &\le H(Y_n | Y_{n-1}, \cdots, Y_1, Y_0, \cdots, Y_{-k}). \\ &\underset{\substack{\uparrow \\ Y_i \text{ : stationary.}}}{=} \underset{\substack{\downarrow \\ H(\mathcal{Y}).}}{H(Y_{n+k+1} | Y_{n+k}, \cdots, Y_1)} \end{aligned}$$

lem 4.5.2.

$H(Y_n | Y_{n-1}, \cdots, Y_1) - H(Y_n | Y_{n-1}, \cdots, Y_1, X_1) \to 0$.

pf). $H(Y_n | Y_{n-1}, \cdots, Y_1) - H(Y_n | Y_{n-1}, \cdots, Y_1, X_1) = I(X_1 ; Y_n | Y_{n-1}, \cdots, Y_1)$.

(Recall $I(X;Y|Z) = I(Y;X|Z) = H(Y|Z) - H(Y|X,Z)$).

Note by chainrule of mutual information,

$$I(X_1;Y_1,\cdots,Y_n) = \sum_{i=1}^{n} I(X_1;Y_i|Y_{i-1},\cdots,Y_1).$$

$\Rightarrow$ $H(X_1) \geq I(X_1;Y_1\cdots,Y_n)$. and $\underline{I(X_1;Y_1\cdots Y_n) \text{ is increasing}}$ with n, hence $\lim_{n\to\infty} I(X_1;Y_1\cdots Y_n)$ exists.

$\left(\because I(X_1;Y_1\cdots,Y_n) - I(X_1;Y_1\cdots Y_{n-1}) = I(X_1;Y_n|Y_1,\cdots,Y_{n-1}) \geq 0\right)$

$$\Rightarrow \lim_{n\to\infty} I(X_1;Y_1,\cdots,Y_n) = \lim_{n\to\infty}\sum_{i=1}^{n} I(X_1;Y_i|Y_{i-1},\cdots,Y_1)$$
$$= \sum_{i=1}^{\infty} I(X_1;Y_i|Y_{i-1},\cdots,Y_1) \text{ converges.}$$

$$\Rightarrow \lim_{n\to\infty} I(X_1;Y_n|Y_1,\cdots,Y_{n-1}) = 0.$$

thm 4.5.1

$\{X_i\}$: stationary Markov chain.

$Y_i = \phi(X_i)$.

$\Rightarrow$ 1. $H(Y_n|Y_{n-1},\cdots,Y_1,X_1) \leq H(\mathcal{Y}) \leq H(Y_n|Y_{n-1},\cdots,Y_1)$.

2. $\lim_{n\to\infty} H(Y_n|Y_{n-1},\cdots,Y_1,X_1) = H(\mathcal{Y}) = \lim_{n\to\infty} H(Y_n|Y_{n-1},\cdots,Y_1)$.

* Hidden Markov model. (HMM).

Y_i is drawn according to $p(y_i|x_i)$, so that,

$$p(x_1,x_2,\cdots,x_n,y_1,\cdots,y_n) = p(x_1)\prod_{i=1}^{n-1} p(x_{i+1}|x_i)\prod_{i=1}^{n} p(y_i|x_i)$$

§. Second law of thermodynamics.

* Second law of thermodynamics : Entropy of an isolated sys. in nondecreasing.

1. Relative entropy $D(\mu_n \| \mu_n')$ decreases with n.

let μ_n, μ_n' : probability dist. of Markov chain at time n
with joint mass ftn p, q resp.

let r : transition probability ftn.

$\Rightarrow$ $p(x_n, x_{n+1}) = p(x_n)\, r(x_{n+1} | x_n)$

$q(x_n, x_{n+1}) = q(x_n)\, r(x_{n+1} | x_n)$

$\Rightarrow$ By chain rule,

$$D(p(x_n, x_{n+1}) \| q(x_n, x_{n+1})) = D(p(x_n) \| q(x_n)) + \overbrace{\boxed{D(p(x_{n+1}|x_n) \| q(x_{n+1}|x_n))}}^{0 \,=\, D(r(x_{n+1}|x_n) \| r(x_{n+1}|x_n)) \,=}$$

$$= D(p(x_{n+1}) \| q(x_{n+1})) + \underbrace{D(p(x_n|x_{n+1}) \| q(x_n|x_{n+1}))}_{\geq 0},$$

$\Rightarrow$ $D(p(x_n) \| q(x_n)) \geq D(p(x_{n+1}) \| q(x_{n+1}))$.

i.e. $D(\mu_n \| \mu_n') \geq D(\mu_{n+1} \| \mu_{n+1}')$.

2. Relative entropy $D(\mu_n \| \mu)$, where μ is stationary dist., decreases with n.

let μ: stationary dist, $\mu_n' = \mu$.
then $\mu_{n+1}' = \mu$.

$\Rightarrow$ By 1. $D(\mu_n \| \mu) \geq D(\mu_{n+1} \| \mu)$.

$\Rightarrow$ $D(\mu_n \| \mu)$ is nonnegative, decreasing, so has a limit.

If μ : unique, then $\lim_{n\to\infty} D(\mu_n \| \mu) = 0$

3. Entropy increases if the stationary dist. is uniform.

Suppose μ: stationary dist, with prob mass ftn u.

$$\Rightarrow D(\mu_n \| \mu) = \sum_{x_n \in \mathcal{X}} p(x_n) \log \frac{p(x_n)}{u(x_n)}.$$

$$= \sum_{x_n \in \mathcal{X}} p(x_n) \log \frac{1}{u(x_n)} - H(X_n).$$

$$= \log |\mathcal{X}| - H(X_n).$$

By 2, $H(X_n) \leq H(X_{n+1})$.

ex). Suppose μ : nonuniform d.
μ' : uniform d.

$X_1 \sim \mu'$

$$\Rightarrow H(X_1) = \log |\mathcal{X}| > H(X_n). \text{ for large } n.$$

$\therefore$ Entropy does not increase

Remark.

Uniform dist is a stationary dist of Markov chain

$\Leftrightarrow$ P is doubly stochastic.

$$\left(\text{i.e. } \sum_i P_{ij} = 1, \ \forall j \quad \sum_j P_{ij} = 1, \ \forall i \right).$$

4. Conditional entropy $H(X_n | X_1)$ increases with n for stationary Markov process.

$\{X_i\}$: stationary Markov process.

$\Rightarrow \quad H(X_n) = -\sum_{x_n \in \mathcal{X}} p(x_n) \log p(x_n).$ is consistent with n.

$$\Rightarrow \quad H(X_n|X_1) \geq H(X_n|X_1, X_2).$$
$$= H(X_n|X_2).$$
$$= H(X_{n-1}|X_1).$$

$\therefore \quad H(X_n|X_1)$ increases with n.